KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Basic Rules

Name _____

S1

Differentiate with respect to x.

$y = 2x^3 + 2x^2 + 2$

S2

Differentiate with respect to x.

$y = 2x^{\frac{4}{3}}$

S3

Differentiate with respect to x.

$y = 4x^{-4}$

S4

Differentiate with respect to x.

$y = 4x^5 + 2x^2 + 5$

Show your working

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ALG-1258-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Basic Rules

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E1

Differentiate with respect to x.

$y = 4x^{\frac{4}{2}}$

E2

Differentiate with respect to x.

$y = -2x^{-5}$

E3

Differentiate with respect to x.

$y = 5x^6 + 4x^2 + 5$

Show your working

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Differentiate with respect to x.

$y = 5x^6 + 4x^2 + 5$

ALG-1258-L1 · E

Show your working

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ALG-1258-L1 · E

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Differentiate with respect to x.

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Differentiate with respect to x.

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ALG-1258-L1 · E

Show your working

Teacher answers & working

Product, Quotient and Chain Rule · Basic Rules

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S1

Differentiate with respect to x.

$y = 2x^3 + 2x^2 + 2$

Answer:

$\frac{dy}{dx} = 6x^2 + 4x$

Working:

Use the Power Rule and show the substitution into the rule.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Combine the term derivatives.

S2

Differentiate with respect to x.

$y = 2x^{\frac{4}{3}}$

Answer:

$\frac{dy}{dx} = \frac{8}{3}x^{\frac{1}{3}}$

Working:

Use the Power Rule and show the substitution into the rule.

Multiply by the fractional power.

Subtract 1 by using a common denominator.

Simplify the coefficient and power.

Write the simplified derivative.

S3

Differentiate with respect to x.

$y = 4x^{-4}$

Answer:

$\frac{dy}{dx} = -16x^{-5}$

Working:

Use the Power Rule and show the substitution into the rule.

Negative powers follow the same power rule.

Write the simplified derivative.

Teacher answers & working

Product, Quotient and Chain Rule · Basic Rules

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S4

Differentiate with respect to x.

$y = 4x^5 + 2x^2 + 5$

Answer:

$\frac{dy}{dx} = 20x^4 + 4x$

Working:

Use the Power Rule and show the substitution into the rule.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Combine the term derivatives.

E1

Differentiate with respect to x.

$y = 4x^{\frac{4}{2}}$

Answer:

$\frac{dy}{dx} = 8x^{\frac{2}{2}}$

Working:

Use the Power Rule and show the substitution into the rule.

Multiply by the fractional power.

Subtract 1 by using a common denominator.

Simplify the coefficient and power.

Write the simplified derivative.

E2

Differentiate with respect to x.

$y = -2x^{-5}$

Answer:

$\frac{dy}{dx} = 10x^{-6}$

Working:

Use the Power Rule and show the substitution into the rule.

Negative powers follow the same power rule.

Write the simplified derivative.

Teacher answers & working

Product, Quotient and Chain Rule · Basic Rules

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E3

Differentiate with respect to x.

$y = 5x^6 + 4x^2 + 5$

Answer:

$\frac{dy}{dx} = 30x^5 + 8x$

Working:

Use the Power Rule and show the substitution into the rule.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Calculate the coefficient and exponent.

Differentiate this term.

Combine the term derivatives.